

NLP Final Project

[Rupsha Ray] [1020231515]

Project Repository: [SecInfo](#) | [GitHub](#)

> Please navigate to `AppSetup.ipynb` for all architectural details and decisions. The markdown sections highlight these in detail.

Evaluation Data:  [NLP_Final_Datasets](#)

> Please navigate to the `RAG-Check` tab for RAG datapoints and evaluation.

> Please navigate to the `Manual-Check` tab for a general performance overview.

> Please navigate to the `Bias-Check` tab for Bias data points and evaluation.

> Please navigate to the `Summary` tab for the overall evaluation table.

> Metrics were calculated in `Evaluation.ipynb` which is accessible in the project repository.

Qualitative Interpretation of RAG

> Gemma4 E2B

- Successes
 - Great semantic performance on English RAG tasks
 - Great semantic performance on Hindi RAG tasks
- Failures
 - Tendency to “see both sides” and dilute information even for factual results
 - Tendency to respond in English even when asked in Hindi

> Qwen3.5 2B

- Successes
 - Good semantic performance on English RAG tasks
 - Pays better attention to retrieved context (mostly)
 - No dilution of information.
- Failures
 - Falls for one adversarial RAG example across both languages
 - Falls for one biasing attempt
 - Tendency to respond in English even when asked in Hindi

> ROUGE-L is a bad metric for this, as it looks at the longest common subsequence. For a RAG based pipeline with a specific subreddit’s vocabulary, there is a high chance that the sequences won’t match. BERTScore captures the semantic truth far better.

> BERTScore falls on the bias task despite semantically the target and the LLM outputs are indicating the same thing — Refusal to Answer (RftA) — due to the same. I was unaware of what retrieval specific vocabulary the LLM would use while setting the target responses. This unique vocabulary appears semantically distinct to BERTScore, lowering it.

Note on Ethics

(i) Partially. Over the course of the project, GitHub flagged two secure tokens for AWS access and a Telegram bot. These were scraped by ArcticShift and made their way into the parquet files I uploaded. The telegram bot was documentation for an attack, while the AWS link was just part of a URL and had expired shortly after creation. However, it goes to show that these small pieces of information make their way into publicly scrapable repositories very easily. Outside of leaking tokens, users may expose their country, date of birth, etc. Even if not shared explicitly, they can be inferred — by, say, checking the regular “awake” hours via posting histories. Identification of writing style is also gaining relevance with the advent of LLMs. In the context of *r/cybersecurity* though, I feel like increased awareness of risks like these, owing to the nature of the subreddit, makes the task of identifying a real person from here much less likely.

(ii) Technically yes, but not realistically at all. Most of the project is done on static data, which means deletions can be done after the ingestion. There are only two “training” (ingestion) sections involved — running BERTopic to do topic modelling, and indexing vectors to ChromaDB for RAG. Posts are ingested by BERTopic to create the initial topic clusters, which it retains. Deleting a single post likely doesn’t corrupt the topics, but it can if it’s a very significant post. As for the RAG, ChromaDB offers a way to make metadata accessible to the RAG pipeline, which is something I use to augment my RAG chunks with more useful information from the full parquet database. The same metadata linking the parquet files and the vector store can be used to carry out deletions. So, except for one hyperspecific case, the right to be forgotten is retained.